Evaluation runs: **prodcons** (producer-consumer-civl)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	3	3	conforms	0.0	4	1 h, 250 GB	3	6	brain03, 2026-11-03
3	4	4	conforms	0.0	4	1 h, 250 GB	4	10	brain03, 2026-11-03
4	5	5	conforms	0.0	4	1 h, 250 GB	5	15	brain03, 2026-11-03
5	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-03
6	7	7	conforms	0.0	4	1 h, 250 GB	7	28	brain03, 2026-11-03
7	8	8	conforms	0.0	4	1 h, 250 GB	8	36	brain03, 2026-11-03
8	9	9	conforms	0.0	4	1 h, 250 GB	9	45	brain03, 2026-11-03
9	10	10	conforms	0.0	4	1 h, 250 GB	10	55	brain03, 2026-11-03
10	11	11	conforms	0.0	4	1 h, 250 GB	11	66	brain03, 2026-11-03

stateless: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	3	8	5	violation	0.0	4	1 h, 250 GB	8	21	brain03, 2026-11-03
3	4	24	20	violation	0.0	4	1 h, 250 GB	24	90	brain03, 2026-11-03
4	5	89	84	violation	0.1	2	1 h, 250 GB	89	435	brain03, 2026-11-03
5	6	415	409	violation	0.2	6	1 h, 250 GB	415	2475	brain03, 2026-11-03
6	7	2372	2365	violation	0.6	20	1 h, 250 GB	2372	16583	brain03, 2026-11-03
7	8	16072	16064	violation	3.7	128	1 h, 250 GB	16072	128548	brain03, 2026-11-03
8	9	125673	125664	violation	245.0	1086	1 h, 250 GB	125673	1131021	brain03, 2026-11-03
9	10			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 20 fifo.
skipped: a smaller size had reached its limit: 10 bag.

stateful

stateful: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	3	3	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-03
3	4	4	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	5	5	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-03
5	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
6	7	7	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-03
7	8	8	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-03
8	9	9	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-03
9	10	10	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-03
10	11	11	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03

stateful: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	3	8	5	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-03
3	4	24	20	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	5	89	84	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
5	6	415	409	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-03
6	7	2372	2365	violation	0.2	16	1 h, 250 GB	brain02, 2026-11-03
7	8	16072	16064	violation	2.3	142	1 h, 250 GB	brain02, 2026-11-03
8	9	125673	125664	violation	374.6	1249	1 h, 250 GB	brain02, 2026-11-03
9				(1 h)			1 h, 250 GB	brain02, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 20 fifo.
skipped: a smaller size had reached its limit: 10 bag.

offline

offline: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	3	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	4	4	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
4	5	5	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
5	6	6	conforms	0.2	6	4 h, 250 GB	0.2	0.2	0.0	merge	brain02, 2026-11-03
6	7	7	conforms	0.1	8	4 h, 250 GB	0.1	0.1	0.0	merge	brain02, 2026-11-03
7	8	8	conforms	0.2	8	4 h, 250 GB	0.2	0.1	0.0	merge	brain02, 2026-11-03
8	9	9	conforms	0.1	8	4 h, 250 GB	0.1	0.1	0.0	merge	brain02, 2026-11-03
9	10	10	conforms	0.1	8	4 h, 250 GB	0.1	0.1	0.0	merge	brain02, 2026-11-03
10	11	11	conforms	0.1	8	4 h, 250 GB	0.1	0.1	0.0	merge	brain02, 2026-11-03

offline: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	8	5	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-0
3	4	24	20	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-0
4	5	89	84	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-0
5	6	415	409	violation	0.4	6	4 h, 250 GB	0.2	0.3	0.1	merge	brain02, 2026-11-0
6	7	2372	2365	violation	0.3	6	4 h, 250 GB	0.1	0.2	0.1	merge	brain02, 2026-11-0
7	8	16072	16064	violation	0.9	24	4 h, 250 GB	0.1	0.5	0.4	merge	brain02, 2026-11-0
8	9	125673	125664	violation	7.2	172	4 h, 250 GB	0.1	2.4	4.7	merge	brain02, 2026-11-0
9	10	1112083	1112073	violation	105.8	1625	4 h, 250 GB	0.1	62.1	43.7	merge	brain02, 2026-11-0
10	11	10976184	10976173	violation	583.0	75383	1 h, 250 GB	0.1	134.3	448.7	in-memory	brain05, 2026-11-0

Not run (offline). planned but never scheduled (grid points beyond the ladder): 11 bag, 12 bag, 20 fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	3	conforms	4.8	158	4 h, 250 GB	0.2	4.8	0.0	merge	brain02, 2026-11-03
3	4	4	conforms	3.1	164	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
4	5	5	conforms	3.1	174	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	8	5	violation	10.3	298	4 h, 250 GB	0.1	10.3	0.1	merge	brain02, 2026-11-03
3	4	24	20	violation	3.4	168	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain01, 2026-11-02
4	5	89	84	violation	7.2	167	10 min, 16 GB	0.1	3.3	3.9	in-memory	brain01, 2026-11-02
5	6	415	409	violation	4.6	191	48 h, 200 GB	0.0	4.6	0.0	merge	volta05, 2026-11-07

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 5 fifo.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	3	conforms	9.8	168	4 h, 250 GB	9.8	0.2	0.0	merge	brain02, 2026-11-03
3	4	4	conforms	4.2	164	4 h, 250 GB	4.2	0.1	0.0	merge	brain02, 2026-11-03
4	5	5	conforms	4.5	165	4 h, 250 GB	4.5	0.4	0.0	merge	brain02, 2026-11-03
5	6	6	conforms	5.4	174	48 h, 80 GB	5.4	0.3	0.0	merge	pinky17, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	8	5	violation	9.4	164	4 h, 250 GB	9.4	0.1	0.0	merge	brain02, 2026-11-03
3	4	24	20	violation	9.3	156	4 h, 250 GB	9.3	0.1	0.0	merge	brain02, 2026-11-03
4	5	89	84	violation	9.8	168	4 h, 250 GB	9.8	0.1	0.0	merge	brain02, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	3	conforms	0.7	147	1 h, 250 GB		brain05, 2026-11-02
3	4	4	conforms	0.6	146	1 h, 250 GB		brain05, 2026-11-02
4	5	5	conforms	0.6	147	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	3	8	5	violation	0.6	148	1 h, 250 GB	62	brain05, 2026-11-02
3	4	24	20	violation	0.7	148	1 h, 250 GB	61	brain05, 2026-11-02
4	5	89	84	violation	1.0	148	1 h, 250 GB	84	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 fifo, 2 bag, 3 fifo, 3 bag, 4 bag, 4 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	3	conforms	3.1	158	1 h, 250 GB		brain04, 2026-11-01
3	4	4	conforms	3.4	158	1 h, 250 GB		brain04, 2026-11-01
4	5	5	conforms	2.9	160	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **single**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	3	8	5	violation	3.1	154	1 h, 250 GB	1	brain04, 2026-11-01
3	4	24	20	violation	3.1	168	1 h, 250 GB	1	brain04, 2026-11-01
4	5	89	84	violation	3.4	163	1 h, 250 GB	2	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): 2 fifo, 2 bag, 3 fifo, 3 bag, 4 bag, 4 fifo.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	6	6	conforms*	9.0	174	48 h, 80 GB	8.9	0.3	0.0	merge	pinky17, 2026-11-07

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	8	conforms	3.8	160	4 h, 250 GB	3.8	0.1	0.0	merge	brain02, 2026-11-03
3	24	24	conforms	4.6	157	10 min, 250 GB	4.6	0.1	0.0	merge	brain02, 2026-11-03
4	89	89	conforms	4.8	157	10 min, 250 GB	4.7	0.1	0.1	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	3	conforms	10.4	814	4 h, 250 GB	10.4	0.2	0.0	merge	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	4	4	conforms	4.2	160	10 min, 250 GB	4.2	0.1	0.0	merge	brain02, 2026-11-03
4	5	5	conforms	4.5	164	10 min, 250 GB	4.5	0.4	0.0	merge	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 bag.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	6	6	conforms*	8.9	174	48 h, 80 GB	0.3	8.9	0.0	merge	pinky17, 2026-11-07

context-free twins offline, Java port against the Rust implementation: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	8	conforms	3.8	160	4 h, 250 GB	0.1	3.8	0.0	merge	brain02, 2026-11-03
3	24	24	conforms	4.6	157	10 min, 250 GB	0.1	4.6	0.0	merge	brain02, 2026-11-03
4	89	89	conforms	4.7	157	10 min, 250 GB	0.1	4.7	0.0	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	3	conforms	10.4	814	4 h, 250 GB	0.2	10.4	0.0	merge	brain02, 2026-11-03
3	4	4	conforms	4.2	160	10 min, 250 GB	0.1	4.2	0.0	merge	brain02, 2026-11-03
4	5	5	conforms	4.5	164	10 min, 250 GB	0.4	4.5	0.0	merge	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 5 bag.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	8	conforms	3.3	152	1 h, 250 GB		brain02, 2026-11-03
3	24	24	conforms	4.6	163	1 h, 250 GB		brain02, 2026-11-03
4	89	89	conforms	5.1	165	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	3	conforms	3.3	157	1 h, 250 GB		brain02, 2026-11-03
3	4	4	conforms	4.5	160	1 h, 250 GB		brain02, 2026-11-03
4	5	5	conforms	4.8	166	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	8	conforms	3.1	156	1 h, 250 GB		brain02, 2026-11-03
3	24	24	conforms	3.1	154	1 h, 250 GB		brain02, 2026-11-03
4	89	89	conforms	3.8	176	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	3	conforms	4.8	160	1 h, 250 GB		brain02, 2026-11-03
3	4	4	conforms	3.1	156	1 h, 250 GB		brain02, 2026-11-03
4	5	5	conforms	3.1	174	1 h, 250 GB		brain02, 2026-11-03